

SAAD MUHAMMAD KHAN

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EDUCATION

National University of Science & Technology (NUST)

Expected September 2028

Bachelor of Computer Science

EXPERIENCE

Independent Software Developer

From August 2024

- Worked on **Machine Learning and Deep Learning**, including **CNNs**, **BERT-based transformer models**, data preprocessing, normalization, feature engineering, and model evaluation. Worked on **Full-Stack Web Development** using **React.js**, **Node.js**, and **Express.js**, including **RESTful APIs**, database systems, and data retrieval. Worked on **Application Development** using **Flutter** and **Java** with backend API integration and database connectivity.

Full Stack Engineer Intern | AptechMedia, Peshawar

June 2026 – August 2026

- Currently working on **full-stack web development and software engineering**, contributing to web applications, backend APIs, database functionality, and client requirements

Java Programming Virtual Intern | CodeAlpha

January 2025 – February 2025

- Completed a virtual internship focused on **object-oriented programming (OOP)**, **problem-solving**, and **Java-based application development**

PROJECTS

Sports Analytics Website (Web Application) | MERN Stack | Tailwind CSS

- Built a full-stack sports analytics platform using **React**, **Node.js**, **Express.js**, and **MongoDB**, with **RESTful APIs** and database schemas for match statistics and player metrics (**17,000+** sample records), tested across **10+ API routes** using **Postman**
- Developed a **React-based analytics dashboard** for interactive visualization of sports performance data and trends

Bankchain | Modular Blockchain Banking System | Solidity, Hardhat, Ethers.js, React

- Developed a **modular Ethereum-based banking system** using **Solidity**, **Hardhat**, and **OpenZeppelin**, with a **multi-contract architecture** for deposits, withdrawals, transfers, loan lifecycle, and card management
- Integrated **Ethers.js** with **React** for wallet interaction and smart contract communication; implemented **on-chain audit logging**, **Ownable**, **ReentrancyGuard**, and a **Hardhat workflow** for deployment and testing

Image Mosaic to Hyperspectral Imaging (HSI) Reconstruction | Python | PyTorch | CNNs

- Developed a **deep learning pipeline** using **CNN-based spectral reconstruction models** to reconstruct **61-band hyperspectral cubes** from low-cost mosaic inputs for the **Hyper-Object Challenge**
- Implemented preprocessing for **normalization**, **tensor reshaping**, and **spectral band alignment**, processing **1024×1024×61 hyperspectral datasets** for multi-band spectral cube generation

BERT-Based Multilingual Polarization Detection Model | Python | PyTorch | Transformers | NLP

- Built a **multilingual text polarization detection system** across English, Chinese, Urdu, Arabic, and Turkish datasets; fine-tuned **RoBERTa-Large** and **XLM-RoBERTa-Large**
- Used **PyTorch** and **Hugging Face Transformers** with multilingual tokenization, subword encoding, and **class-weighted cross-entropy loss**; uploaded trained models to the **Hugging Face Hub**

CLIENT PROJECTS

Dental Clinic Website (Web Application) | MERN Stack | Tailwind CSS

- Developed a full-stack clinic platform with a **public-facing website** and **admin dashboard**, featuring patient booking and dynamic doctor management

Pen & Page Academia (Web Application) | MERN Stack | Tailwind CSS

- Developed a responsive full-stack website for the institute using the **MERN stack**, showcasing its **programs**, **services**, and **academic information**

TECH STACK

Frontend Development: HTML5, CSS3, JavaScript (ES6+), React.js, Tailwind CSS

Backend Development: Node.js, Express.js, Flask, JDBC, RESTful API Development, Authentication Systems

Machine Learning: Python (NumPy, Pandas, Scikit-learn), PyTorch, TensorFlow, Transformers, NLP, CNN

Mobile Development: Flutter, Java, Responsive UI/UX Design

Databases: PostgreSQL, MySQL, Oracle Database, MongoDB, Firebase

DevOps & Tools: Git, GitHub, Postman, VS Code

AI Tools: Claude, Cursor, Antigravity, Gemini, Codex, ChatGPT

Programming Languages: Python, JavaScript, C++, Java, Dart, Assembly Language (AVR, MASM)

ACADEMIC PERFORMANCE

Maintained consistent academic excellence with a CGPA of **3.76 / 4.00**, demonstrating strong performance in core computer science subjects and analytical problem-solving skills.